

Problem set 5

Part I: Rolling two dice

1. Write a function called `double_six` that takes a number `n` as an argument, simulates rolling two fair dice `n` times, and returns `TRUE` if at least one of the `n` rolls is a double-six.

Hint: generate two vectors of rolls and check whether any pair equals (6, 6).

```
double_six <- function(n){  
  die1 <- sample(1:6, n, replace = TRUE)  
  die2 <- sample(1:6, n, replace = TRUE)  
  
  any(die1 == 6 & die2 == 6)  
}
```

2. Suppose you roll two fair dice 24 times. If we define success to mean seeing at least one double-six, what is the estimated probability of success? Use a Monte Carlo simulation with `B = 1000` trials based on the function `double_six` from the previous exercise.

```
B <- 10^3  
n <- 24  
  
results <- replicate(B, double_six(n))  
mean(results)
```

3. Redo the previous exercise for several values of `n` to determine at what number of rolls the probability becomes greater than 50%. Set the seed at 1997.

```
set.seed(1997)  
  
compute_prob <- function(n, B = 10^3){  
  mean(replicate(B, double_six(n)))  
}
```

```
n_vals <- 1:60
probs <- sapply(n_vals, compute_prob)

which(probs > 0.5)[1]
```

4. These probabilities can be computed exactly instead of relying on Monte Carlo approximations. Since the probability of not getting a double-six on a single roll is $35/36$, we have

$$\Pr(\text{at least one double-six in } n \text{ rolls}) = 1 - \left(\frac{35}{36}\right)^n.$$

Plot the probabilities you obtained using Monte Carlo as points and the exact probabilities with a red line.

```
exact_prob <- function(n){
  1 - (35/36)^n
}

exact_vals <- exact_prob(n_vals)

plot(n_vals, probs, pch = 16, col = "blue",
     xlab = "n", ylab = "Probability")

lines(n_vals, exact_vals, col = "red", lwd = 2)
```

5. The Monte Carlo points in question 4 will not exactly match the red curve because the simulation uses only 1,000 iterations. Repeat exercise 2 for $n = 24$, but now try $B \leftarrow \text{seq}(10, 250, 5)^2$ iterations. Plot the estimated probability against $\sqrt{B}$. At what value of $\sqrt{B}$ do the estimates consistently stay within 0.005 of the exact probability? Add horizontal lines at the exact probability plus and minus 0.005. Set the seed to 1998.

```
set.seed(1998)

B <- seq(10, 250, 5)^2
n <- 24

estimates <- sapply(B, function(b){
  mean(replicate(b, double_six(n)))
})
```

```

exact <- 1 - (35/36)^n

plot(sqrt(B), estimates, type = "b",
     xlab = "sqrt(B)", ylab = "Estimated Probability")

abline(h = exact, col = "red", lwd = 2)
abline(h = exact + 0.005, col = "blue", lty = 2)
abline(h = exact - 0.005, col = "blue", lty = 2)

```

6. Repeat the comparison from question 4 (Monte Carlo points versus exact red line), but now use your findings from question 5 to choose an appropriate value of B so that the points practically fall on the red curve.

Hint: choose a value of B that is large enough for a good plot, but not so large that your document takes too long to render.

```

n_vals <- 1:60
B <- 15000

probs_largeB <- sapply(n_vals, function(n){
  mean(replicate(B, double_six(n)))
})

exact_vals <- 1 - (35/36)^n_vals

plot(n_vals, probs_largeB, pch = 16, col = "blue",
     xlab = "n", ylab = "Probability")

lines(n_vals, exact_vals, col = "red", lwd = 2)

```

Part II: American roulette

In American roulette there are 38 slots total: 18 red, 18 black, and 2 green.

7. If a player bets \$1 on red, what is the probability that the casino wins the bet?

$$P(\text{casino wins}) = P(\text{not red}) = \frac{18 + 2}{38} = \frac{20}{38} = \frac{10}{19}.$$

8. If a player bets \$1 on red, the casino's profit from that single bet is represented by the random variable X :

The random variable X represents the casino's profit from a single bet: [$X =$

If the player bets on red: - If the ball lands on red (18 out of 38 outcomes), the player wins \$1, so the casino loses \$1, so $X = -1$. - If the ball lands on black or green (20 out of 38 outcomes), the player loses \$1, so the casino gains \$1, so $X = +1$.

$$X = \begin{cases} -1 & \text{with probability } \frac{18}{38} \\ +1 & \text{with probability } \frac{20}{38} \end{cases}$$

]

Create a sampling model for X using the `sample` function.

```
X <- function(n){  
  sample(c(-1, 1), size = n, replace = TRUE,  
         prob = c(18/38, 20/38))  
}
```

9. Now create a random variable S for the casino's total profit if $n = 1000$ people each make a \$1 bet on red. Use Monte Carlo simulation with $B = 10000$ trials to estimate the probability that the casino loses money.

```
n <- 1000  
B <- 10^4  
  
# function for one experiment: total casino profit from n players  
S <- function(n){  
  sum(sample(c(-1, 1), size = n, replace = TRUE,  
            prob = c(18/38, 20/38)))  
}  
  
# simulate B experiments  
S_vals <- replicate(B, S(n))  
  
# estimate probability casino loses money (S < 0)  
mean(S_vals < 0)
```

10. What is the expected value of X ?

$$E[X] = (-1) \cdot \frac{18}{38} + (1) \cdot \frac{20}{38} = \frac{20 - 18}{38} = \frac{2}{38} = \frac{1}{19}.$$

11. What is the standard error of X ?

$$E[X^2] = 1^2 \cdot \frac{18}{38} + 1^2 \cdot \frac{20}{38} = 1$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 1 - \left(\frac{1}{19}\right)^2 = 1 - \frac{1}{361} = \frac{360}{361}$$

$$SE(X) = \sqrt{\frac{360}{361}} \approx 0.999$$

12. What is the expected value of S ? Does the Monte Carlo simulation confirm this?

$$E[S] = nE[X] = 1000 \cdot \frac{1}{19} = \frac{1000}{19} \approx 52.63.$$

```
mean(S_vals)
```

13. What is the standard error of S ? Does the Monte Carlo simulation confirm this?

$$\text{Var}(S) = n \cdot \text{Var}(X) = 1000 \cdot \frac{360}{361}$$

$$SE(S) = \sqrt{1000 \cdot \frac{360}{361}} \approx 31.6$$

```
sd(S_vals)
```

14. Use data visualization to convince yourself that the distribution of S is approximately normal. Make a histogram and a QQ-plot of the standardized values of S . The QQ-plot should be close to the identity line.

```
# standardize S
S_std <- (S_vals - mean(S_vals)) / sd(S_vals)

# Histogram
hist(S_std,
     breaks = 40,
     main = "Histogram of Standardized S",
     xlab = "Standardized S",
     col = "lightblue",
     border = "white")
```

```
# QQ plot
qqnorm(S_std,
      main = "QQ Plot of Standardized S")
qqline(S_std, col = "red", lwd = 2)
```

15. The normal approximation will not be perfect in the tails. What would improve the approximation more: increasing the number of people playing n , or increasing the number of Monte Carlo iterations B ? Explain your reasoning.

Increasing n improves the normal approximation because the central limit theorem depends on having a larger sample size, which makes the distribution S closer to normal. Increasing B only improves the accuracy of the simulation results, but it does not change the shape of the distribution itself.

$$S \approx N\left(\mu = \frac{1000}{19}, \sigma = \sqrt{1000 \cdot \frac{360}{361}}\right)$$

$$P(S < 0) = P\left(Z < \frac{0 - \mu}{\sigma}\right) = P\left(Z < \frac{-1000/19}{\sqrt{1000 \cdot 360/361}}\right)$$

16. Use the Central Limit Theorem to approximate the probability that the casino loses money. Does your CLT approximation agree with the Monte Carlo estimate from question 9?

$$P(S < 0) \approx 0.01$$

Using the CLT:

$$S \approx N\left(\mu = \frac{n}{19}, \sigma = \sqrt{n \cdot \frac{360}{361}}\right)$$

$$P(S < 0) = P\left(Z < \frac{0 - n/19}{\sqrt{n \cdot 360/361}}\right)$$

Set this equal to 0.01:

$$\frac{-n/19}{\sqrt{n \cdot 360/361}} \approx -2.33$$

Solve numerically for (n) .

```

target_prob <- 0.01

find_prob <- function(n){
  mu <- n * (1/19)
  sigma <- sqrt(n * (360/361))
  pnorm(0, mean = mu, sd = sigma)
}

n_vals <- 1:10000
probs <- sapply(n_vals, find_prob)

n_min <- n_vals[which(probs <= target_prob)[1]]
n_min

```

17. What is the minimum number of people n who must each bet \$1 on red so that the probability the casino loses money is approximately 1% according to the CLT? Check your answer with a Monte Carlo simulation.

$$X = \begin{cases} -1 & \text{with probability } \frac{18}{38} \\ +1 & \text{with probability } \frac{20}{38} \end{cases}$$

$$\mu = E[X] = \frac{1}{19}, \quad \sigma^2 = \frac{360}{361}$$

$$S \approx N\left(\frac{n}{19}, \sqrt{n \cdot \frac{360}{361}}\right)$$

We want:

$$P(S < 0) = 0.01$$

Standardizing:

$$P(S < 0) = P\left(Z < \frac{0 - \frac{n}{19}}{\sqrt{n \cdot \frac{360}{361}}}\right)$$

Set equal to the 1% quantile:

$$\frac{-n/19}{\sqrt{n \cdot \frac{360}{361}}} = -2.33$$

Simplify:

$$\frac{n/19}{\sqrt{n} \cdot \sqrt{360/361}} = 2.33$$

$$\frac{\sqrt{n}}{19\sqrt{360/361}} = 2.33$$

Solve for (n):

$$\sqrt{n} = 2.33 \cdot 19 \cdot \sqrt{\frac{360}{361}}$$

$$n = \left(2.33 \cdot 19 \cdot \sqrt{\frac{360}{361}} \right)^2$$

$$n \approx 1950$$

18.

```
set.seed(1999)

# Recompute n_min (so this chunk works independently)
target_prob <- 0.01

find_prob <- function(n){
  mu <- n * (1/19)
  sigma <- sqrt(n * (360/361))
  pnorm(0, mean = mu, sd = sigma)
}

n_vals <- 1:10000
probs <- sapply(n_vals, find_prob)
n_min <- n_vals[which(probs <= target_prob)[1]]

# Monte Carlo verification
B <- 10^4

S <- function(n){
  sum(sample(c(-1, 1), size = n, replace = TRUE,
             prob = c(18/38, 20/38)))
}
```



```
S_vals <- replicate(B, S(n_min))  
mean(S_vals < 0)
```